Installation of Apache Tomcat and Jenkins Pipeline for MyMavenWebApp01

Installation of Apache Tomcat:

Step 1: To install Apache tomcat in root folder, type the command **sudo -i** and then **sudo apt update && sudo upgrade -y**

```
shreya@ShreyaM:~$ sudo -i
[sudo] password for shreya:
root@ShreyaM:~# sudo apt update && sudo upgrade -y
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudo: upgrade: command not found
```

Enter tmp by typing the command **cd /tmp** and download Apache Tomcat software by typing the command **wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz**

```
root@ShreyaM:~# cd /tmp
root@ShreyaM:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-14 20:17:58--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: ‘apache-tomcat-9.0.95.tar.gz’

apache-tomcat-9.0.9 100%[===================>]  12.13M   156KB/s    in 70s

2026-04-14 20:19:10 (177 KB/s) - ‘apache-tomcat-9.0.95.tar.gz’ saved [12715996/12715996]
```

Then enter type **cd ..** and extract the downloaded to opt in root by typing the commands

cd opt

mkdir tomcat

cd ..

cd tmp

to extract the tar file into tomcat folder run the command **sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1**

```
root@ShreyaM:/tmp# cd ..
root@ShreyaM:/# cd opt
root@ShreyaM:/opt# ls
google
root@ShreyaM:/opt# mkdir tomcat
root@ShreyaM:/opt# cd ..
root@ShreyaM:/# cd tmp
```

```
root@ShreyaM:/tmp# sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1
apache-tomcat-9.0.95/conf/
apache-tomcat-9.0.95/conf/catalina.policy
apache-tomcat-9.0.95/conf/catalina.properties
apache-tomcat-9.0.95/conf/context.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xsd
apache-tomcat-9.0.95/conf/logging.properties
apache-tomcat-9.0.95/conf/server.xml
apache-tomcat-9.0.95/conf/tomcat-users.xml
apache-tomcat-9.0.95/conf/tomcat-users.xsd
apache-tomcat-9.0.95/conf/web.xml
apache-tomcat-9.0.95/bin/
apache-tomcat-9.0.95/lib/
apache-tomcat-9.0.95/logs/
apache-tomcat-9.0.95/temp/
apache-tomcat-9.0.95/webapps/
apache-tomcat-9.0.95/webapps/ROOT/
apache-tomcat-9.0.95/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.95/webapps/docs/
apache-tomcat-9.0.95/webapps/docs/META-INF/
apache-tomcat-9.0.95/webapps/docs/WEB-INF/
apache-tomcat-9.0.95/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.95/webapps/docs/annotationapi/
apache-tomcat-9.0.95/webapps/docs/api/
apache-tomcat-9.0.95/webapps/docs/appdev/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/web/images/
```

Enter tomcat and type **ls** to see all files are extracted

Enter conf **cd conf** and type ls to see server.

```
apache-tomcat-9.0.95/bin/makebase.sh
apache-tomcat-9.0.95/bin/setclasspath.sh
apache-tomcat-9.0.95/bin/shutdown.sh
apache-tomcat-9.0.95/bin/startup.sh
apache-tomcat-9.0.95/bin/tool-wrapper.sh
apache-tomcat-9.0.95/bin/version.sh
root@ShreyaM:/tmp# cd ..
root@ShreyaM:/# cd opt
root@ShreyaM:/opt# cd tomcat/
root@ShreyaM:/opt/tomcat# ls
bin           CONTRIBUTING.md  logs       RELEASE-NOTES  webapps
BUILDING.txt  lib              NOTICE     RUNNING.txt    work
conf          LICENSE          README.md  temp
root@ShreyaM:/opt/tomcat# cd conf
root@ShreyaM:/opt/tomcat/conf# ls
catalina.policy      jaspic-providers.xml  server.xml        web.xml
catalina.properties  jaspic-providers.xsd  tomcat-users.xml
context.xml          logging.properties    tomcat-users.xsd
```

xml use this file to change the port number from default 8080 to any number say 9090 because Jenkins is running on 8080

```
root@ShreyaM:/opt/tomcat/conf# gedit server.xml

(gedit:10897): dconf-WARNING **: 20:22:41.590: failed to commit changes to dconf: Failed to execute child process “dbus-launch” (No such file or
ry)

(gedit:10897): dconf-WARNING **: 20:22:41.616: failed to commit changes to dconf: Failed to execute child process “dbus-launch” (No such file or
ry)

(gedit:10897): dconf-WARNING **: 20:22:57.634: failed to commit changes to dconf: Failed to execute child process “dbus-launch” (No such file or
```

Type **gedit server.xml** and go to line number 71 and change the port number to and save

```
-->
<Connector port="9090" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           maxParameterCount="1000"
           />
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
           port="8080" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           maxParameterCount="1000"
           />
```

we need to create tomcat users as well so add the users in tomcat-users.xml file by typing the command **gedit tomcat-users.xml**

go to line 22 add the line **<user username="admin" password="Admin" roles="manager-gui"/>**

```
<tomcat-users xmlns="http://tomcat.apache.org/xml"
              xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
              xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
              version="1.0">
              <user username="admin" password="Admin" roles="manager-gui"/>
<!--
```

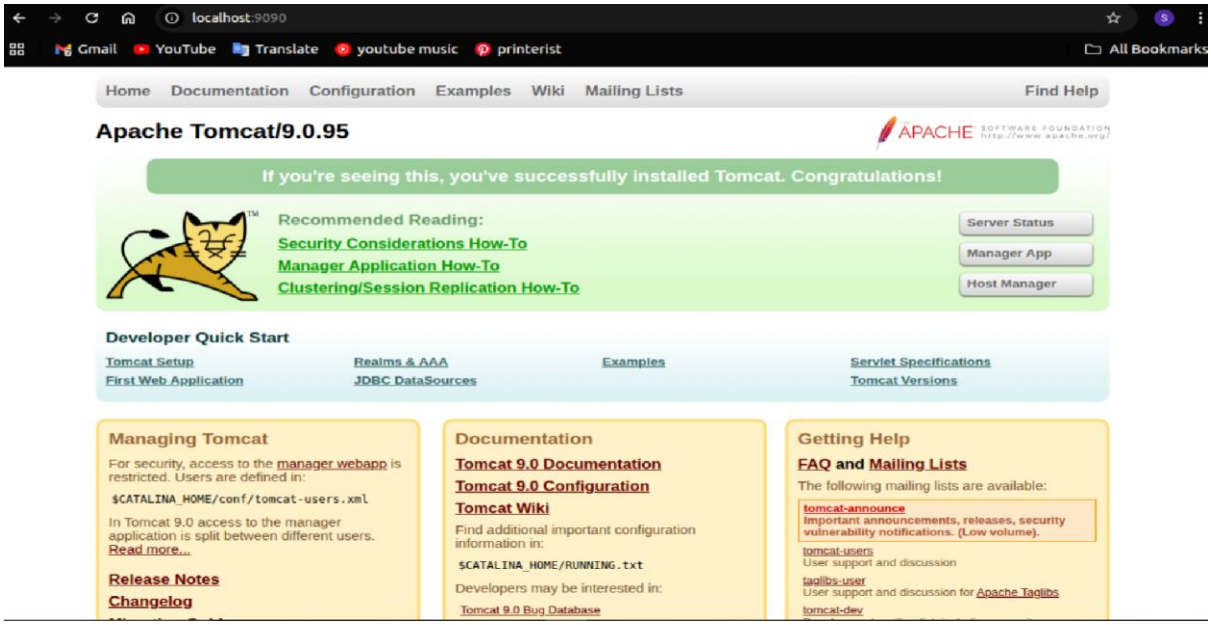
Then enter bin and type **ls**

```
root@ShreyaM:/opt/tomcat/conf# cd ..
root@ShreyaM:/opt/tomcat# cd bin
root@ShreyaM:/opt/tomcat/bin# ls
bootstrap.jar                configtest.sh     shutdown.sh
catalina.bat                 daemon.sh         startup.bat
catalina.sh                  digest.bat        startup.sh
catalina-tasks.xml           digest.sh         tomcat-juli.jar
ciphers.bat                  makebase.bat      tomcat-native.tar.gz
ciphers.sh                   makebase.sh       tool-wrapper.bat
commons-daemon.jar           setclasspath.bat  tool-wrapper.sh
commons-daemon-native.tar.gz setclasspath.sh   version.bat
configtest.bat               shutdown.bat      version.sh
```

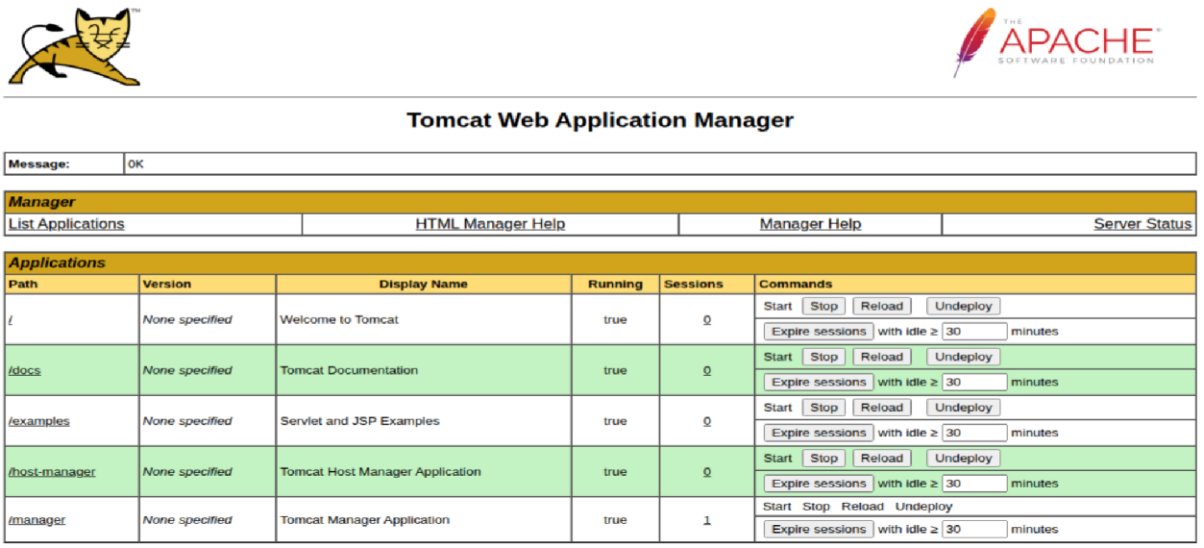
we can type the commands **startup.sh** and **shutdown.sh** to start the tomcat and stop the tomcat

```
root@ShreyaM:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ShreyaM:/opt/tomcat/bin#
```

open the page in browser say localhost:9090



Click on Manager App enter the credentials previously provided



Jenkins Pipeline for MymavenWebApp

Step 1: to create MymavenWebApp:

mvn archetype:generate -DgroupId=com.example -DartifactId=MymavenWebApp -DarchetypeArtifactId=maven-archetype-webapp -DinterativeMode=false

```
shreya@ShreyaM:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MymavenWebApp -DarchetypeArtifactId=maven-archetype-webapp -Dinteract
iveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MymavenWebApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/devops/MymavenWebApp
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  4.118 s
[INFO] Finished at: 2026-04-08T08:38:04+05:30
```

Step 2: open the terminal and type the command **cd MymavenWebApp** and

Go to index.jsp file and edit the hello wold code and paste the following code

```
shreya@ShreyaM:~/devops/MymavenWebApp$ cd src/main/webapp
shreya@ShreyaM:~/devops/MymavenWebApp/src/main/webapp$ gedit index.jsp
```

```
<html>
<body>
<h2>Hello World! name :shreya mehtry usn :1BI23CS199</h2>
</body>
</html>
```

Got to pom.xml and edit :

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MymavenWebApp</artifactId>
  <packaging>war</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MymavenWebApp Maven Webapp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <!-- Servlet API for Java EE -->
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>javax.servlet-api</artifactId>
            <version>4.0.1</version>
            <scope>provided</scope>
        </dependency>
  </dependencies>
  <build>
  <plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
```

```
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version> <!-- Update to latest stable version -->
</plugin>
</plugins>
    <finalName>MymavenWebApp</finalName>
  </build>
</project>
```

Mvn clean install:

```
shreya@ShreyaM:~/devops/MymavenWebApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:MymavenWebApp >-----------------------
[INFO] Building MymavenWebApp Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/2.2/maven-war-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/2.2/maven-war-plugin-2.2.pom (6.5 kB at 7.1 kB/s
)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/2.2/maven-war-plugin-2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/2.2/maven-war-plugin-2.2.jar (79 kB at 338 kB/s)
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MymavenWebApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MymavenWebApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MymavenWebApp ---
```

```
shreya@ShreyaM:~/devops/MymavenWebApp$ tree
.
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MymavenWebApp01
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MymavenWebApp01.war

12 directories, 7 files
shreya@ShreyaM:~/devops/MymavenWebApp$
```

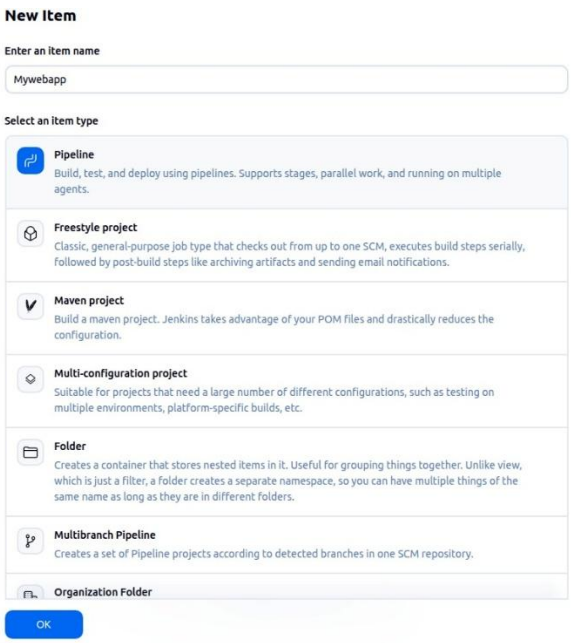
Step 3: create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

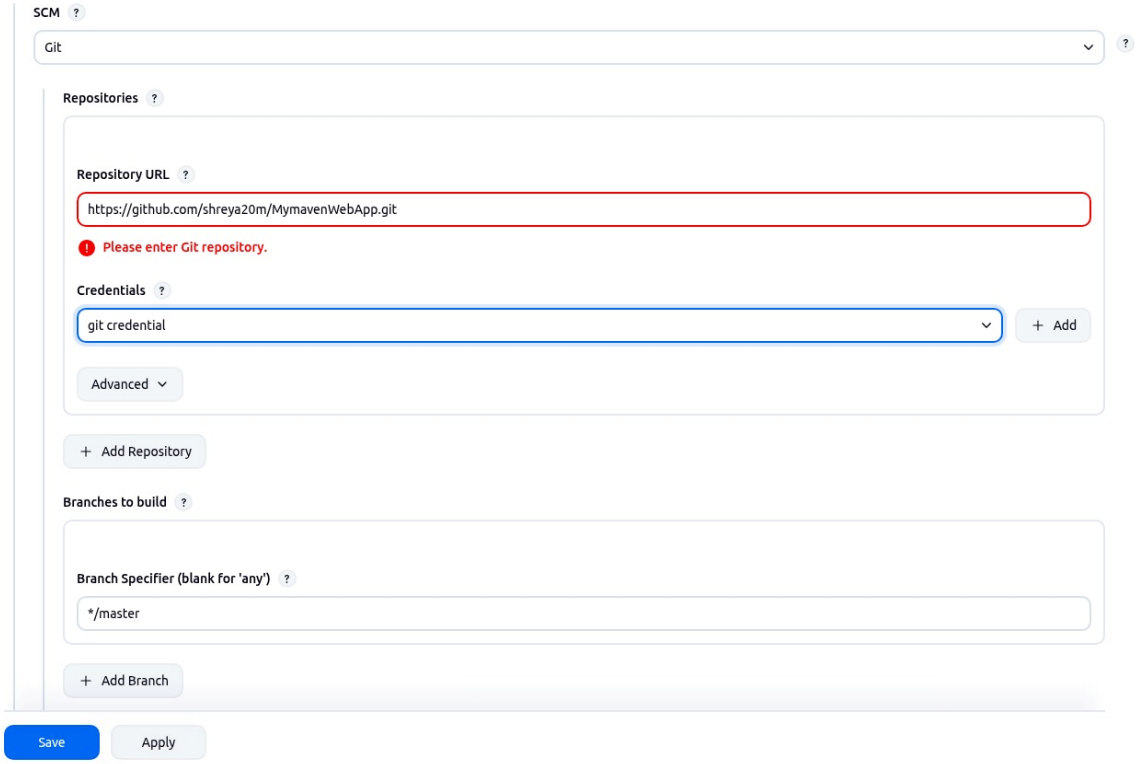
```
pipeline {
    agent any  // Use any available agent


    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
```

```
stages {
    stage('Checkout') {
        steps {
            git branch: 'master', url: ' https://github.com/shreya20m/MymavenWebApp.git'
        }}
    stage('Build') {
        steps {
            sh 'mvn clean package'  // Run Maven build
        }}
    stage('Test') {
        steps {
            sh 'mvn test'  // Run unit tests
        }}
    stage('Deploy WAR') {
        steps {
            sh 'cp target/MymavenWebApp.war /opt/tomcat/webapps/'
        }}}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }}}
```

Step 4: Go to Jenkis Dashboard click on new item enter the item name as MyMavenWebApp and select pipeline. In the configure select pipeline scm paste the repository url select github credentials applyand save





Step 5: go **MyMavenWebApp** in Jenkins and click **Build Now**; the build completes successfully.

Mywebapp - Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Deploy WAR	Declarative: Post Actions
Average stage times: (full run time: ~3min 30s)	4s	1s	6s	1min 16s	23s	4s	1s
#1 Apr 15 09:07 No Changes	4s	1s	6s	1min 16s	23s	4s	1s

Step 6: go to Tomcat Web Application Manager and select the path /MyavenWebApp

Tomcat Web Application Manager

Message:	OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy

localhost:9090/MymavenWebApp/

Hello World! name :shreya mehtry usn :1BI23CS199

Devops